

LINQ, Delegates și expresii Lambda Laborator 4

Tehnici avansate de programare

Nicoleta Scrimint



Email course for SOLID

- <https://mailchi.mp/nimblepros/solid-email-course>



Developer Events I

- Iasi
 - <https://devdays.ro/>
 - <https://codecamp.ro/codecamps/iasi-2026/>
- Bucharest
 - <https://www.gotech.world/>



Developer Events II

- Community
 - <https://www.dotnetconf.net/>
 - <https://ignite.microsoft.com/en-US/home>
 - <https://ndcoslo.com/>
 - <https://qconlondon.com/>
 - <https://www.thoughtworks.com/about-us/events/xconf>
 - <https://2026.dddeurope.com/>
 - <https://gotocph.com/2026>



Videos

- <https://learn.microsoft.com/en-us/shows/csharp-for-beginners/linq-method-syntax-vs-query-csharp-for-beginners>
- <https://learn.microsoft.com/en-us/shows/csharp-for-beginners/linq-query-expressions-from-where-orderby-and-select-csharp-for-beginners>
- <https://learn.microsoft.com/en-us/shows/csharp-for-beginners/language-integrated-query-linq-and-ienumerable-csharp-for-beginners>



Learning sources

- <https://learn.microsoft.com/en-us/dotnet/csharp/programming-guide/delegates/>
- <https://learn.microsoft.com/en-us/dotnet/csharp/language-reference/operators/lambda-expressions>
- <https://learn.microsoft.com/en-us/dotnet/standard/linq/>
- <https://learn.microsoft.com/en-us/dotnet/csharp/linq/>
- <https://learn.microsoft.com/en-us/dotnet/csharp/linq/get-started/introduction-to-linq-queries>
- <https://learn.microsoft.com/en-us/dotnet/csharp/linq/get-started/query-expression-basics>



Exercises

- <https://learn.microsoft.com/en-us/dotnet/csharp/tutorials/working-with-linq>
- <https://learn.microsoft.com/en-us/dotnet/csharp/linq/how-to-extend-linq>



Mandatory homework II

- Functional requirement: ProductCatalog search
 - Follow SOLID principles
 - Get products with filters, order, groups
 - Take into account performance when retrieving the query results
- Technical requirements
 - Use Lambda expressions
 - Use Linq query methods
 - Use Linq query language
- Deadline: 16 March 2026 end of the day